# Dalyn Wachala

Mechatronics Engineering Student

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## PROFESSIONAL SUMMARY

MTSU Mechatronics Engineering Junior and aspiring controls engineer with a focused interest in robotic systems and controls. Combines hands-on experience in collaborative robotics environments with a passion for engineering research. Dedicated to pursuing advanced academic study to solve the next generation of complex mechatronic challenges.

## EDUCATION

**Middle Tennessee State University | 3.825 GPA**

- B.S. in Mechatronics Engineering with a minor in Mathematics | August 2023 - May 2027
- Recipient of Presidential Scholarship from the school. Dean's list for 4 out of 5 semesters.

## WORK EXPERIENCE

### Teaching Assistant

MTSU | January 2026 - Present

- Run weekly sessions for a class of 60 students, walking through tough problems in equilibrium and structural analysis to help them learn the material.
- Tutor students one-on-one and in groups, helping them work through difficult topics like truss analysis, friction, and internal forces.
- Grade assignments and help choose practice problems for the class, making sure the work stays at the high standard expected for a core engineering course.

### Robotics Research Assistant

MTSU | December 2025 - Present

- Developing an Unmanned Aerial Vehicle (UAV) platform from the ground up, integrating high-performance motors, ESCs, and communication hardware for autonomous flight research.
- Configuring and programming Pixhawk flight controllers using Linux-based environments to implement custom flight parameters and advanced motor control algorithms.
- Executing precision hardware integration through professional soldering and electrical troubleshooting to ensure reliable system communication between transmitters and receivers.

### Process Engineering Intern

Valeo | January 2025 - January 2026

- Optimized assembly line performance by diagnosing and repairing complex mechanical and electrical faults to minimize downtime.
- Analyzed and improved line efficiency through data-driven process adjustments, contributing to increased throughput and operational reliability.
- Collaborated on the design and implementation of new assembly lines, assisting in the technical setup and integration of part-specific production systems.

## PROJECTS

**3lb Combat Robot: Vertical Spinner**

- Acted as Team Captain for a 3lb vertical spinner project, managing the team through the design and build phases.
- Managed the full electrical design, including the wiring and motor control setup for both the drive and weapon systems.
- Selected and integrated key parts like the battery, ESCs, and a safety kill switch to meet competition requirements.

**Micromouse Robot**

- Designed and 3D printed the chassis and internal mounts to keep the robot lightweight and compact.
- Designed a custom PCB to house the electronics and simplify the wiring for the sensors and controllers.
- Programmed the logic for autonomous maze-solving and wall-following to help the robot reach the goal quickly.

## SKILLS

- **Software & OS:** Linux (Ubuntu), Unity, Google Sheets, Arduino IDE, SolidWorks.
- **Electronics & Hardware:** Soldering, Motor Control (ESCs), Pixhawk, Circuit Analysis, PCB Design (KiCAD), Sensor Design.
- **Manufacturing:** 3D Printing (Klipper/Ender 3 v2), Manual Assembly, Simple Machining (Drill Press, Welding, and Band Saw).
- **Soft Skills:** Public Speaking, Technical Tutoring, Project Management.

## LEADERSHIP AND INVOLVEMENT

**MTSU Robotics Club | Vice President**

- Lead club operations and coordinate team projects, helping manage a group of students interested in hands-on robotics and competitive engineering.

**Engineering Technology Department | Student Ambassador**

- Serve as a representative for the department, giving tours of the engineering building to high schoolers and their families to show them what the program is about.
- Focus on public speaking and communication, answering questions about the curriculum and sharing my personal experience as a Mechatronics student.